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(54) **AIR CONDITIONING APPLIANCE AND  
METHODS FOR DETERMINING MAKE-UP  
AIR ACTIONS**

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(57) **ABSTRACT**

An air conditioning appliance or method may include detecting a first indoor pollutant level within an indoor environment while holding an auxiliary fan in an inactive state. The appliance or method may further include activating the auxiliary fan to urge the flow of make-up air to the indoor portion following detecting the first indoor pollutant level and detecting a second indoor pollutant level within the indoor environment following detecting the first indoor pollutant level and activating the auxiliary fan. The appliance or method may still further include determining an outdoor pollution level or a room volume of the indoor environment based on the first and second indoor pollutant levels and estimating a high-pollutant timepoint for the indoor environment based on the determined outdoor pollution level or room volume. The appliance or method may yet further include adjusting activation of the auxiliary fan prior to reaching the high-pollutant timepoint.

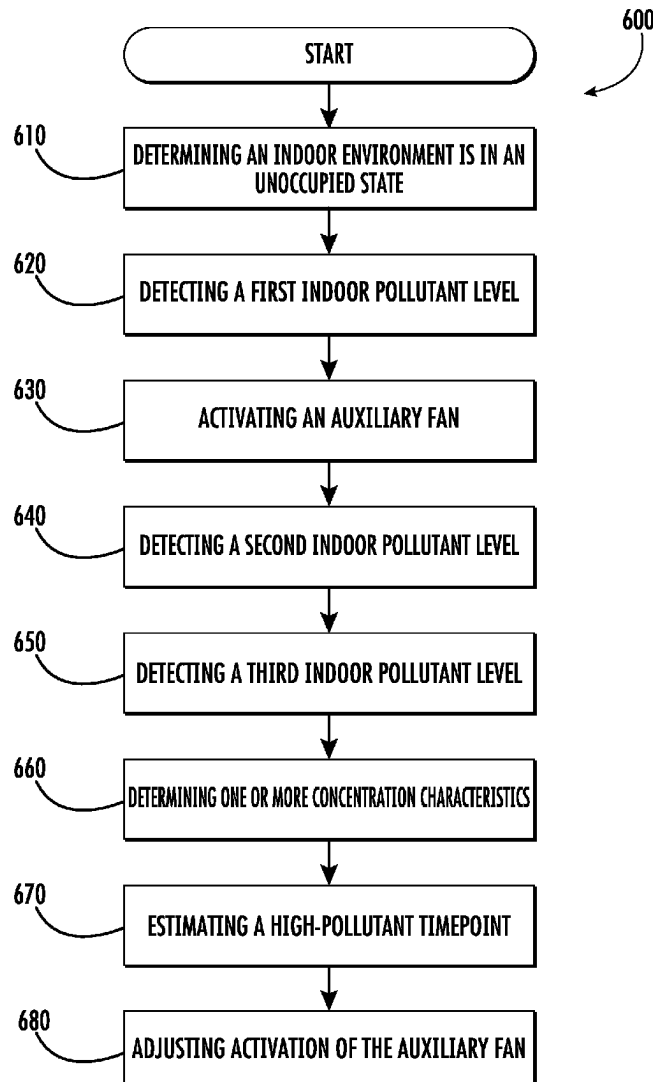
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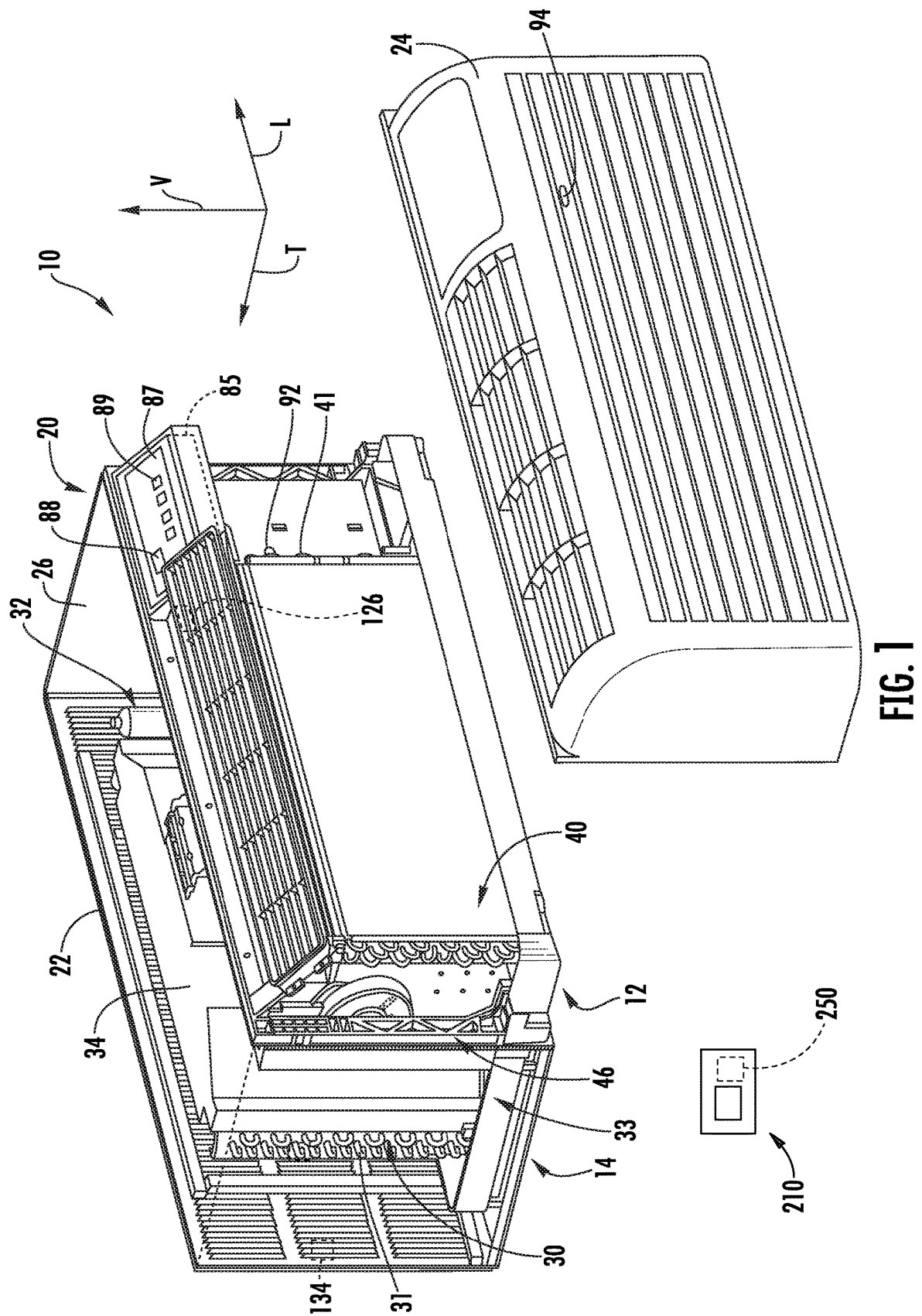
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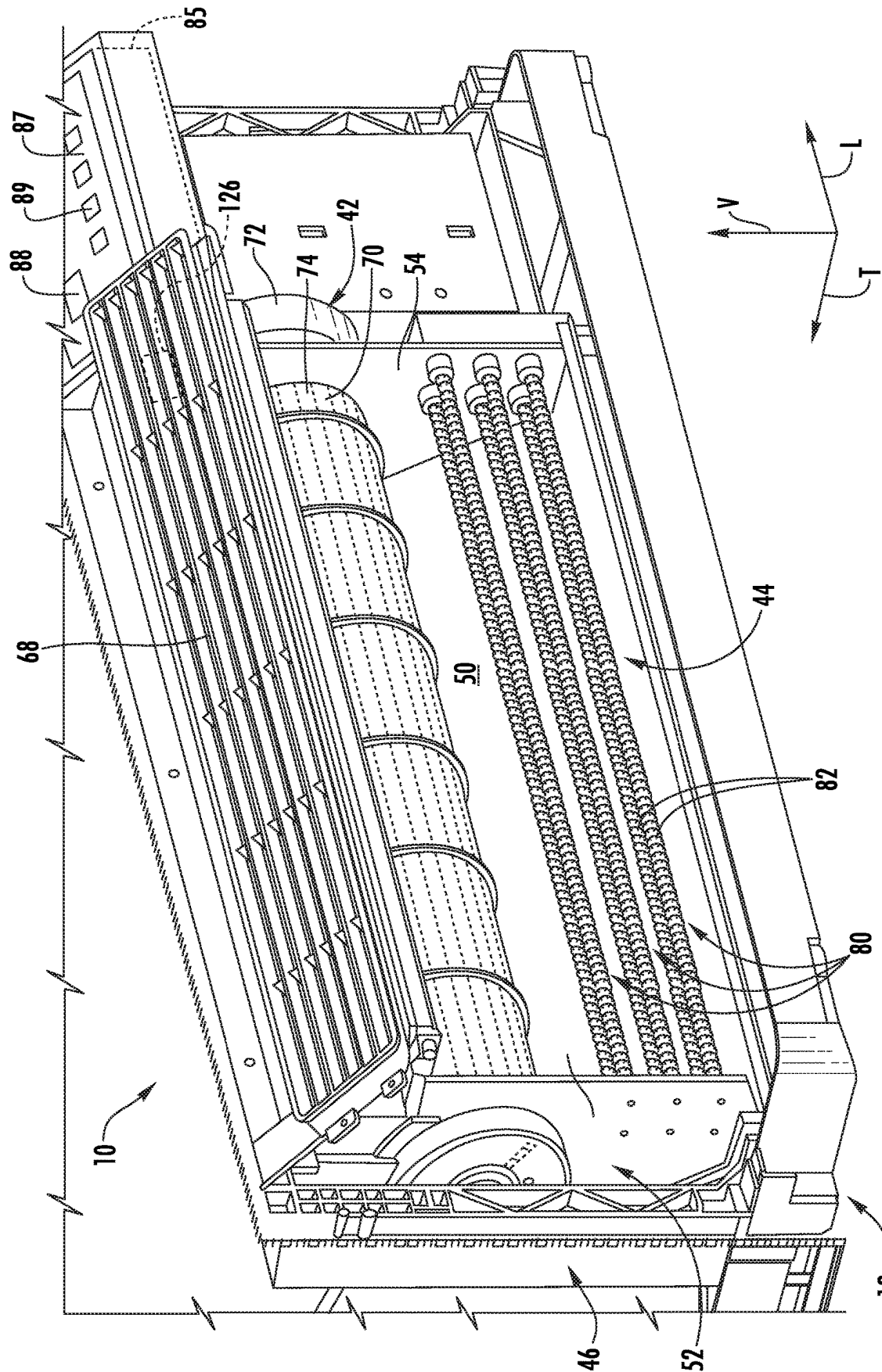


FIG. 2

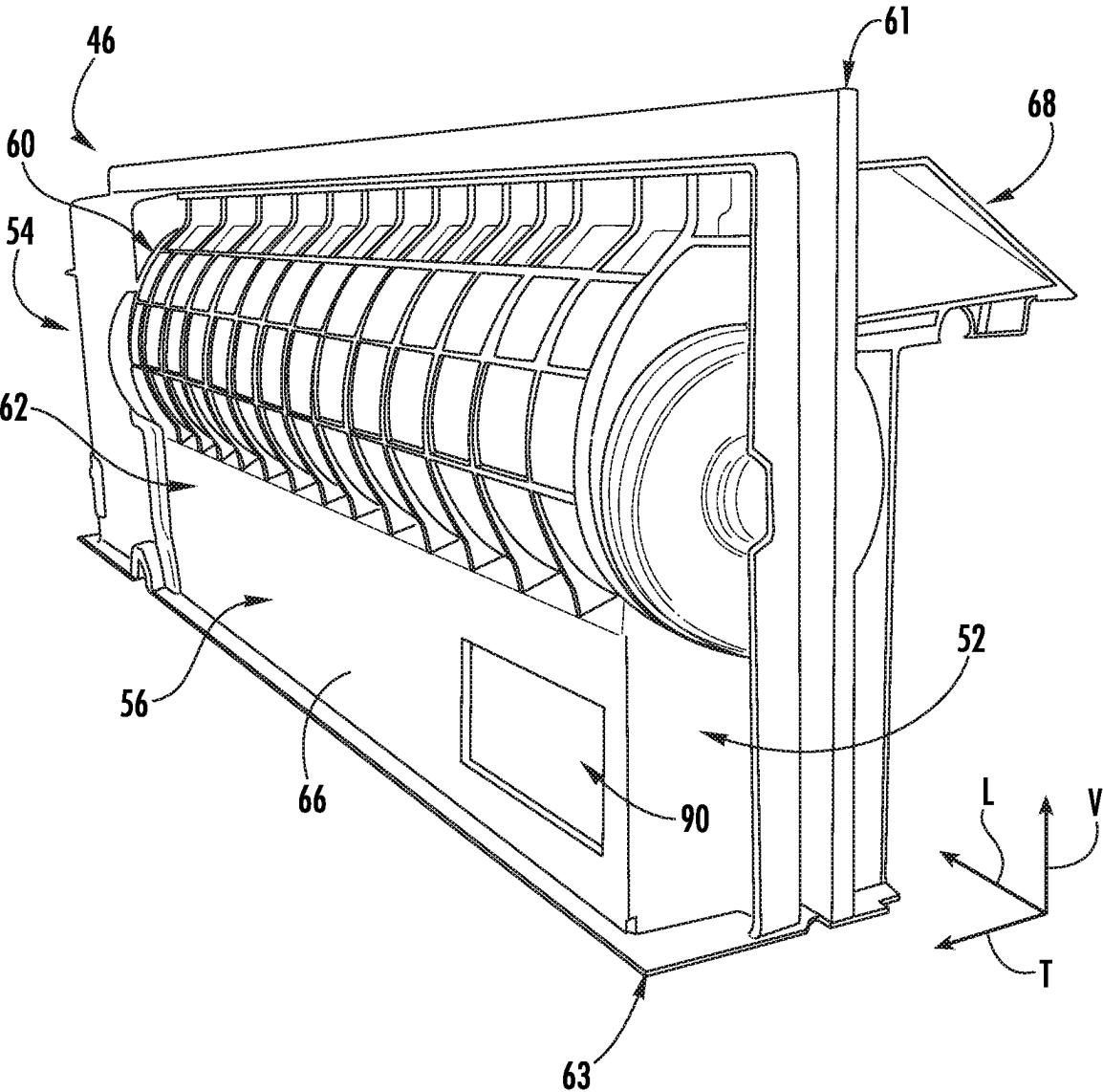
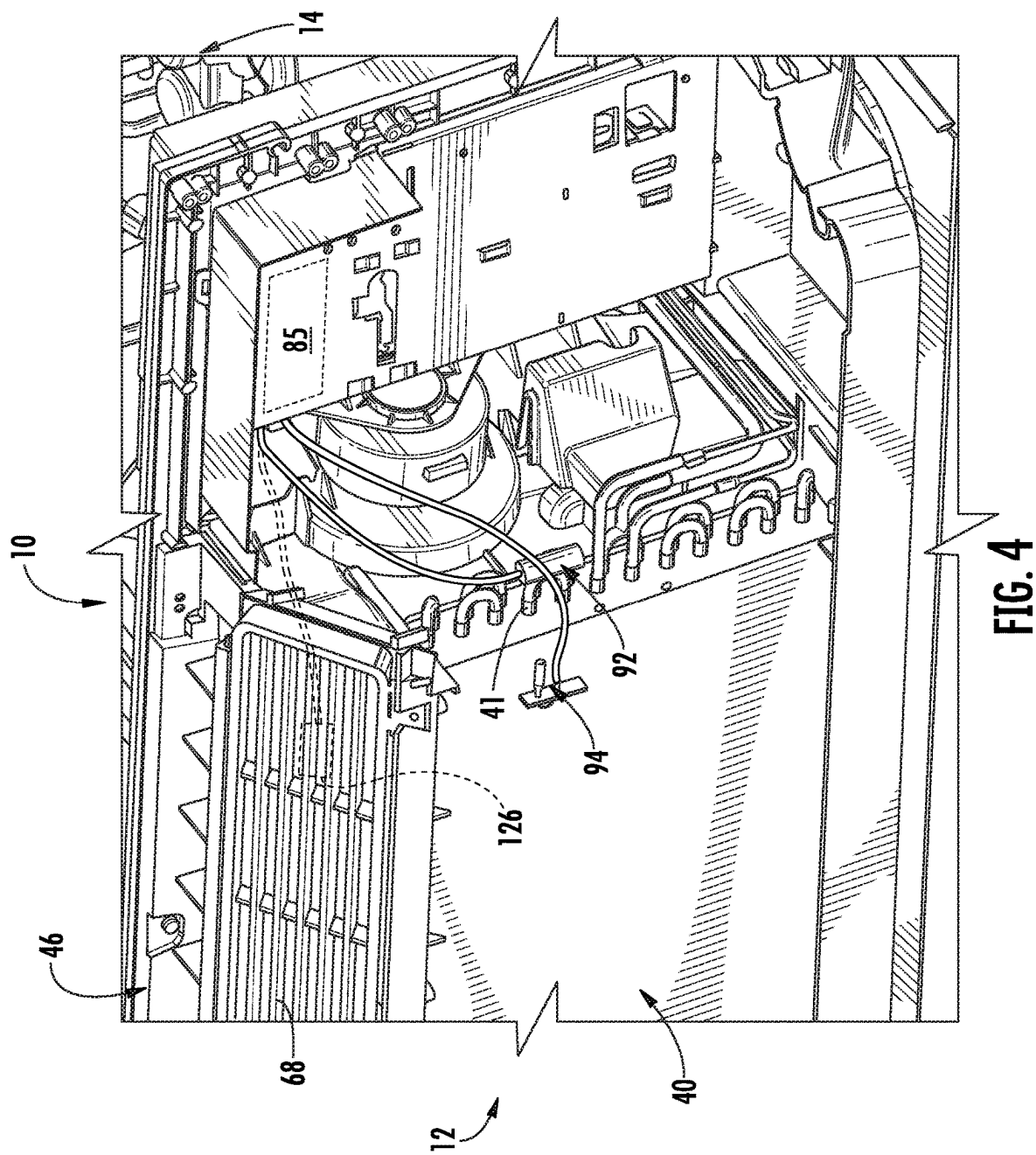


FIG. 3



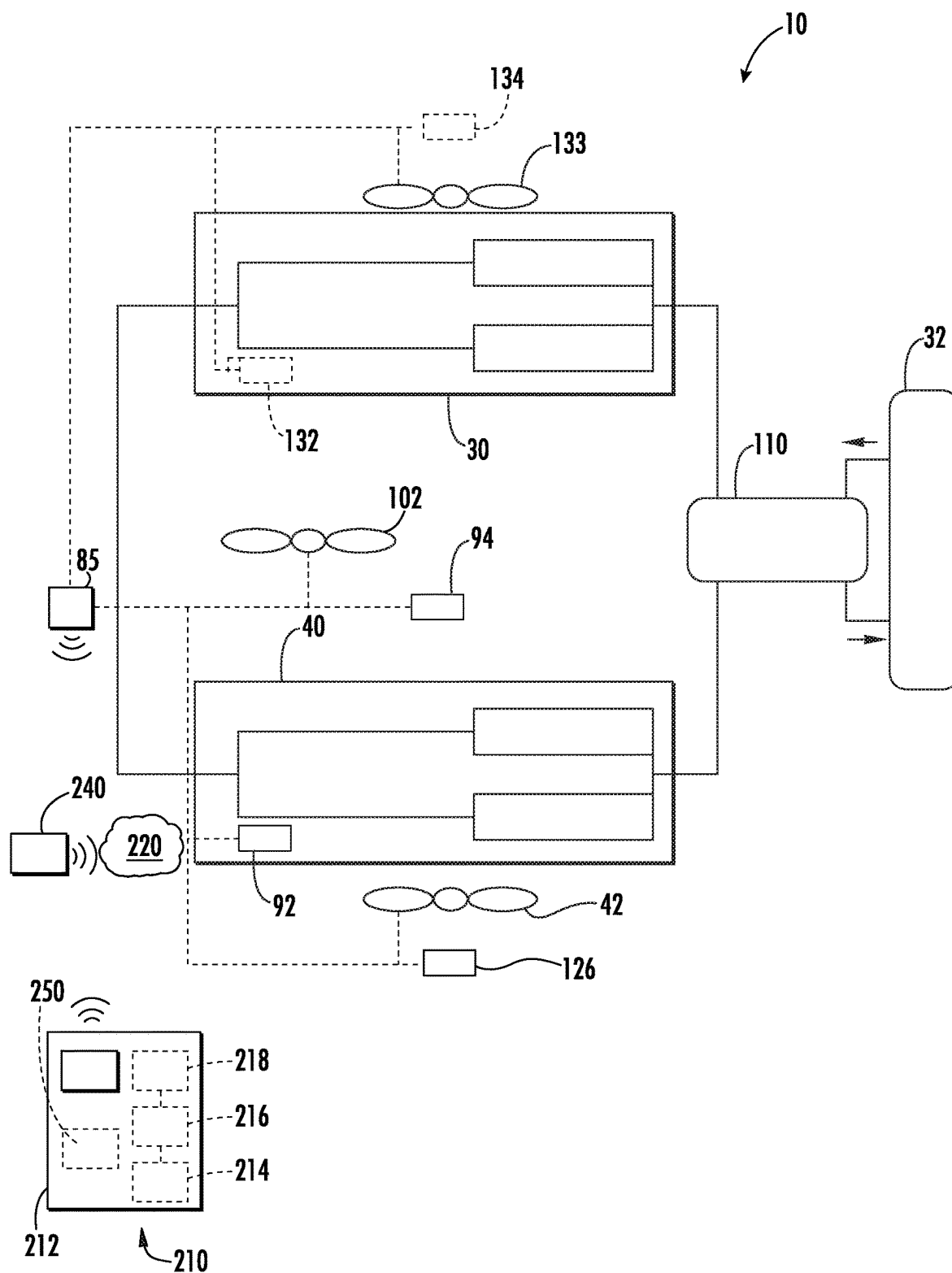
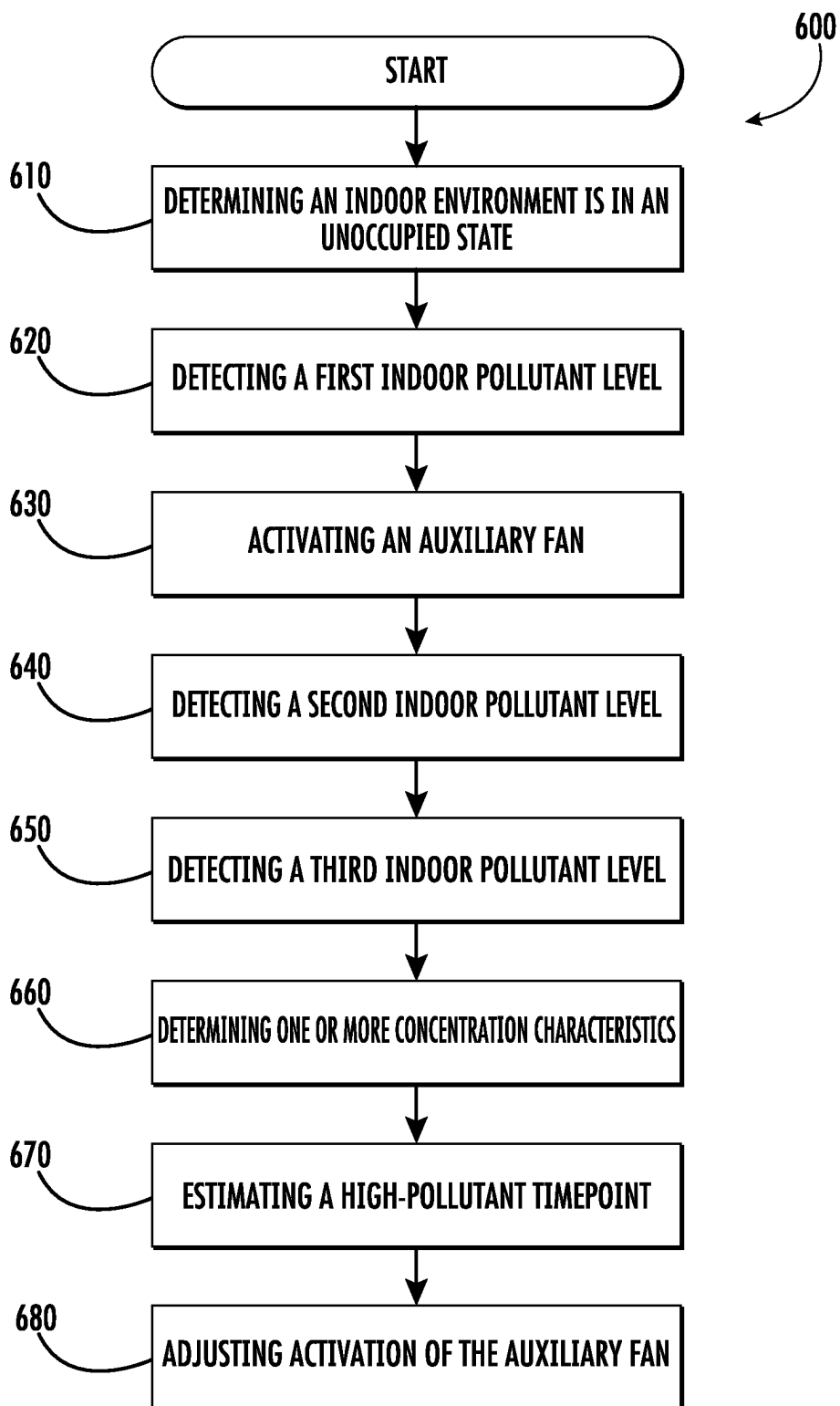


FIG. 5



**FIG. 6**

## AIR CONDITIONING APPLIANCE AND METHODS FOR DETERMINING MAKE-UP AIR ACTIONS

### FIELD OF THE DISCLOSURE

[0001] The present subject matter relates generally to single-package air conditioner units, including methods of operating such units in a manner that accounts for the environment at or in which a unit is installed.

### BACKGROUND OF THE DISCLOSURE

[0002] Air conditioner units are conventionally utilized to adjust the temperature within structures such as dwellings and office buildings. In particular, one-unit type or single-package room air conditioner units, such as window units, single-package vertical units (SPVU), vertical packaged air conditioners (VPAC), or package terminal air conditioners (PTAC) may be utilized to adjust the temperature in, for example, a single room or group of rooms of a structure. Such units are especially common in hotels, rental apartments, and assisted-living facilities in which a large number of occupants live within the same building.

[0003] A typical one-unit type air conditioner or air conditioning appliance includes an indoor portion and an outdoor portion. The indoor portion generally communicates (e.g., exchanges air) with the area within a building, and the outdoor portion generally communicates (e.g., exchanges air) with the area outside a building. Accordingly, the air conditioner unit generally extends through, for example, a wall of the structure. Generally, a fan may be operable to rotate to motivate air through the indoor portion. Another fan may be operable to rotate to motivate air through the outdoor portion. A sealed cooling system including a compressor is generally housed within the air conditioner unit to treat (e.g., cool or heat) air as it is circulated through, for example, the indoor portion of the air conditioner unit. One or more control boards are typically provided to direct the operation of various elements of the particular air conditioner unit.

[0004] Frequently, the indoor space may need to draw in air from the outdoors (i.e., make-up air). For example, if a vent fan is turned on in a bathroom or air is otherwise ejected from the indoor space, fresh air from the outdoors is required. Depending on, for example, the efficiency of the weather stripping around doors and windows, some make-up air could simply be drawn into the indoors by cracks or other openings. If such cracks are not sufficient, the flow of make-up air may be insufficient or too slow. Furthermore, government regulations, such as fire codes may require that cracks or openings be eliminated as much as possible, precluding a sufficient flow of make-up air.

[0005] Accordingly, certain air conditioner units allow for the introduction of make-up air into an indoor space or environment. Unfortunately, conditions may arise in which make-up air would create additional issues. For instance, the outdoor air may contain undesirably high levels of pollutants (e.g., carbon dioxide), such as might be caused by a nearby wildfire or industrial processes. Such conditions might be especially difficult to address when the indoor space is unoccupied, and the unit would otherwise automatically supply make-up air to the room without considering the cumulative affects of the make-up air. Given the wide variety (e.g., size or circulation) of indoor spaces, it may be

especially difficult for a unit or a user to be able to account for the particular conditions of air or the indoor space.

[0006] As a result, it may be useful to provide an air conditioner unit or method of operation capable of addressing one or more of the above-identified issues.

### BRIEF DESCRIPTION OF THE DISCLOSURE

[0007] Aspects and advantages of the invention will be set forth in part in the following description, or may be obvious from the description, or may be learned through practice of the invention.

[0008] In one exemplary aspect of the present disclosure, a method of operating a single-package air conditioner unit is provided. The method may include detecting a first indoor pollutant level within an indoor environment while holding an auxiliary fan in an inactive state. The method may further include activating the auxiliary fan to urge the flow of make-up air to the indoor portion following detecting the first indoor pollutant level and detecting a second indoor pollutant level within the indoor environment following detecting the first indoor pollutant level and activating the auxiliary fan. The method may still further include determining an outdoor pollutant level based on the first and second indoor pollutant levels and estimating a high-pollutant timepoint for the indoor environment based on the determined outdoor pollutant level. The method may yet further include adjusting activation of the auxiliary fan prior to reaching the high-pollutant timepoint.

[0009] In another exemplary aspect of the present disclosure, a method of operating a single-package air conditioner unit is provided. The method may include detecting a first indoor pollutant level within an indoor environment while holding an auxiliary fan in an inactive state. The method may further include activating the auxiliary fan to urge the flow of make-up air to the indoor portion following detecting the first indoor pollutant level and detecting a second indoor pollutant level within the indoor environment following detecting the first indoor pollutant level and activating the auxiliary fan. The method may still further include determining a room volume of the indoor environment based on the first and second indoor pollutant levels and estimating a high-pollutant timepoint for the indoor environment based on the determined room volume. The method may yet further include adjusting activation of the auxiliary fan prior to reaching the high-pollutant timepoint.

[0010] In yet another exemplary aspect of the present disclosure, a single-package air conditioner unit is provided. The single-package air conditioner unit may include a cabinet, an outdoor heat exchanger, an indoor heat exchanger, a compressor, a bulkhead, an auxiliary fan, and a controller. The cabinet may define an outdoor portion and an indoor portion. The outdoor heat exchanger may be disposed in the outdoor portion. The indoor heat exchanger may be disposed in the indoor portion. The compressor may be in fluid communication with the outdoor heat exchanger and the indoor heat exchanger to circulate a refrigerant between the outdoor heat exchanger and the indoor heat exchanger. The bulkhead may be disposed between the indoor portion and the outdoor portion. The bulkhead may define a vent aperture extending therethrough to permit airflow between the indoor portion and the outdoor portion. The auxiliary fan may be mounted within the cabinet in fluid communication with the vent aperture for urging a flow of make-up air from the outdoor portion through the vent



aperture to the indoor portion. The controller may be in operative communication with the compressor and the auxiliary fan. The controller may be configured to initiate a conditioning operation. The conditioning operation may include detecting a first indoor pollutant level within the indoor environment while holding the auxiliary fan in an inactive state, activating the auxiliary fan to urge the flow of make-up air to the indoor portion following detecting the first indoor pollutant level, detecting a second indoor pollutant level within the indoor environment following detecting the first indoor pollutant level and activating the auxiliary fan, determining an outdoor pollutant level and a room volume based on the first and second indoor pollutant levels, estimating a high-pollutant timepoint for the indoor environment based on the determined outdoor pollutant level and the room volume, and adjusting activation of the auxiliary fan prior to reaching the high-pollutant timepoint.

**[0011]** These and other features, aspects and advantages of the present invention will become better understood with reference to the following description and appended claims. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention.

#### BRIEF DESCRIPTION OF THE DRAWINGS

**[0012]** A full and enabling disclosure of the present invention, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures.

**[0013]** FIG. 1 provides a perspective view of an air conditioner unit, with a room front exploded from a remainder of the air conditioner unit for illustrative purposes, in accordance with exemplary embodiments of the present disclosure.

**[0014]** FIG. 2 is a perspective view of components of an indoor portion of an air conditioner unit in accordance with exemplary embodiments of the present disclosure.

**[0015]** FIG. 3 is a rear perspective view of a bulkhead assembly in accordance with exemplary embodiments of the present disclosure.

**[0016]** FIG. 4 is another perspective view of components of an indoor portion of an air conditioner unit in accordance with exemplary embodiments of the present disclosure.

**[0017]** FIG. 5 provides a schematic view of an air conditioner unit according to exemplary embodiments of the present disclosure.

**[0018]** FIG. 6 provides a flow chart illustrating a method of operating an air conditioner unit according to exemplary embodiments of the present disclosure.

**[0019]** Repeat use of reference characters in the present specification and drawings is intended to represent the same or analogous features or elements of the present invention.

#### DETAILED DESCRIPTION

**[0020]** Reference now will be made in detail to embodiments of the invention, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope of the invention. For instance, features illustrated or described as

part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.

**[0021]** Reference now will be made in detail to embodiments of the invention, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents. The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” In addition, references to “an embodiment” or “one embodiment” does not necessarily refer to the same embodiment, although it may. Any implementation described herein as “exemplary” or “an embodiment” is not necessarily to be construed as preferred or advantageous over other implementations.

**[0022]** As used herein, the terms “first,” “second,” and “third” may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components. The terms “includes” and “including” are intended to be inclusive in a manner similar to the term “comprising.” Similarly, the term “or” is generally intended to be inclusive (i.e., “A or B” is intended to mean “A or B or both”). The terms “upstream” and “downstream” refer to the relative flow direction with respect to fluid flow in a fluid pathway. For example, “upstream” refers to the flow direction from which the fluid flows, and “downstream” refers to the flow direction to which the fluid flows. In addition, here and throughout the specification and claims, range limitations may be combined or interchanged. Such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise. For example, all ranges disclosed herein are inclusive of the endpoints, and the endpoints are independently combinable with each other. The singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise.

**[0023]** Approximating language, as used herein throughout the specification and claims, may be applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “generally,” “about,” “approximately,” and “substantially,” are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value, or the precision of the methods or machines for constructing or manufacturing the components or systems. For example, the approximating language may refer to being within a 10 percent margin (i.e., including values within ten percent greater or less than the stated value). In this regard, for example, when used in the context of an angle or direction, such terms include within ten degrees greater or less than the stated angle or direction (e.g., “generally vertical” includes forming an angle of up to

ten degrees in any direction, such as, clockwise or counter-clockwise, with the vertical direction V).

**[0024]** Except as explicitly indicated otherwise, recitation of a singular processing element (e.g., “a controller,” “a processor,” “a microprocessor,” etc.) is understood to include more than one processing element. In other words, “a processing element” is generally understood as “one or more processing element.” Furthermore, barring a specific statement to the contrary, any steps or functions recited as being performed by “the processing element” or “said processing element” are generally understood to be capable of being performed by “any one of the one or more processing elements.” Thus, a first step or function performed by “the processing element” may be performed by “any one of the one or more processing elements,” and a second step or function performed by “the processing element” may be performed by “any one of the one or more processing elements by which the first step or function is performed.” Moreover, it is understood that recitation of “the processing element” or “said processing element” performing a plurality of steps or functions does not require that at least one discrete processing element be capable of performing each one of the plurality of steps or functions.

**[0025]** Generally, aspects of the present disclosure may provide an appliance or method in which conditions of an outdoor environment or indoor space may be automatically (e.g., without direct user intervention) may be determined and accounted for. In some such aspects of the present disclosure, the concentration of pollutants (e.g., from the outdoor environment) within an indoor space may be anticipated or otherwise maintained in a desirable level or range.

**[0026]** Referring now to the figures, in FIGS. 1 through 5, an air conditioner 10 according to various exemplary embodiments is provided. The air conditioner 10 is generally a one-unit type air conditioner, also conventionally referred to as a room air conditioner or package terminal air conditioner unit (PTAC). The air conditioner 10 includes an indoor portion 12 and an outdoor portion 14, and defines a vertical direction V, a lateral direction L, and a transverse direction T. Each direction V, L, T is perpendicular to each other, such that an orthogonal coordinate system is generally defined.

**[0027]** Although described in the context of a PTAC, an air conditioner unit as disclosed herein may be provided as a window unit, single-package vertical unit (SPVU), vertical packaged air conditioner (VPAC), or any other suitable single-package air conditioner. The air conditioner 10 is intended only as an exemplary unit and does not otherwise limit the scope of the present disclosure. Thus, it is understood that the present disclosure may be equally applicable to other types of air conditioner units.

**[0028]** Generally, a cabinet 20 of the air conditioner 10 contains various other components of the air conditioner 10. Cabinet 20 may include, for example, a rear grill 22 and a room front 24 that may be spaced apart along the transverse direction T by a wall sleeve 26. The rear grill 22 may be part of the outdoor portion 14, while the room front 24 is part of the indoor portion 12. Components of the outdoor portion 14, such as an outdoor heat exchanger 30, outdoor fan 33 (FIG. 5), and compressor 32 may be housed within the wall sleeve 26. A casing 34 may additionally enclose the outdoor fan 33, as shown.

**[0029]** Referring now also to FIG. 2, indoor portion 12 may include, for example, an indoor heat exchanger 40, a blower fan 42, and a heating unit 44. These components may, for example, be housed behind the room front 24. Additionally, a bulkhead 46 may generally support or house various other components or portions thereof of the indoor portion 12, such as the blower fan 42 and the heating unit 44. Bulkhead 46 may generally separate and define the indoor portion 12 and outdoor portion 14.

**[0030]** Outdoor and indoor heat exchangers 30, 40 may be components of a thermodynamic assembly (i.e., sealed system), which may be operated as a refrigeration assembly (and thus perform a refrigeration cycle) and, in the case of the heat pump unit embodiment, a heat pump (and thus perform a heat pump cycle). Thus, as is understood, exemplary heat pump unit embodiments may be selectively operated perform a refrigeration cycle at certain instances (e.g., while in a cooling mode) and a heat pump cycle at other instances (e.g., while in a heating mode). By contrast, exemplary A/C exclusive unit embodiments may be unable to perform a heat pump cycle (e.g., while in the heating mode), but still perform a refrigeration cycle (e.g., while in a cooling mode).

**[0031]** In optional embodiments, such as exemplary heat pump unit embodiments, the sealed system includes a reversible refrigerant valve 110 (FIG. 5). Reversible refrigerant valve 110 selectively directs compressed refrigerant from compressor 32 to either indoor heat exchanger 40 or outdoor heat exchanger 30. For example, in a cooling mode, reversible refrigerant valve 110 is arranged or configured to direct compressed refrigerant from compressor 32 to outdoor heat exchanger 30. Conversely, in a heating mode, reversible refrigerant valve 110 is arranged or configured to direct compressed refrigerant from compressor 32 to indoor heat exchanger 40. Thus, reversible refrigerant valve 110 permits the sealed system to adjust between the heating mode and the cooling mode, as will be understood by those skilled in the art.

**[0032]** The assembly may, for example, further include compressor 32 and an expansion valve, both of which may be in fluid communication with the heat exchangers 30, 40 to flow refrigerant therethrough, as is generally understood. Optionally, the compressor 32 may be a variable speed compressor or, alternatively, a single speed compressor. When the assembly is operating in a cooling mode, and thus performs a refrigeration cycle, the indoor heat exchanger 40 acts as an evaporator and the outdoor heat exchanger 30 acts as a condenser. In heat pump unit embodiments, when the assembly is operating in a heating mode, and thus performs a heat pump cycle, the indoor heat exchanger 40 acts as a condenser and the outdoor heat exchanger 30 acts as an evaporator. The outdoor and indoor heat exchangers 30, 40 may each include coils 31, 41, as illustrated, through which a refrigerant may flow for heat exchange purposes, as is generally understood.

**[0033]** Bulkhead 46 may include various peripheral surfaces that define an interior 50 thereof. For example, and additionally referring to FIG. 3, bulkhead 46 may include a first sidewall 52 and a second sidewall 54 which are spaced apart from each other along the lateral direction L. A rear wall 56 may extend laterally between the first sidewall 52 and second sidewall 54.

**[0034]** The rear wall 56 may, for example, include an upper portion 60 and a lower portion 62. Upper portion 60

may for example have a generally curvilinear cross-sectional shape, and may accommodate a portion of the blower fan **42** when blower fan **42** is housed within the interior **50**. Lower portion **62** may have a generally linear cross-sectional shape, and may be positioned below upper portion **60** along the vertical direction V. Rear wall **56** may further include an indoor facing surface **64** and an opposing outdoor facing surface. The indoor facing surface **64** may face the interior **50** and indoor portion **12**, and the outdoor facing surface **66** may face the outdoor portion **14**.

**[0035]** Bulkhead **46** may extend between a top end **61** and a bottom end **63** along vertical axis V. Upper portion **60** may, for example, include top end **61**, while lower portion **62** may, for example, include bottom end **63**. Bulkhead **46** may additionally include, for example, an air diverter **68**, which may extend between the sidewalls **52**, **54** along the lateral direction L and through which air may flow.

**[0036]** In exemplary embodiments, blower fan **42** may be a tangential fan. Alternatively, however, any suitable fan type may be utilized. Blower fan **42** may include a blade assembly **70** and a motor **72**. The blade assembly **70**, which may include one or more blades disposed within a fan housing **74**, may be disposed at least partially within the interior **50** of the bulkhead **46**, such as within the upper portion **60**. As shown, blade assembly **70** may for example extend along the lateral direction L between the first sidewall **52** and the second sidewall **54**. The motor **72** may be connected to the blade assembly **70**, such as through the fan housing **74** to the blades via a shaft. Operation of the motor **72** may rotate the blades, thus generally operating the blower fan **42**. Further, in exemplary embodiments, motor **72** may be disposed exterior to the bulkhead **46**. Accordingly, the shaft may for example extend through one of the sidewalls **52**, **54** to connect the motor **72** and blade assembly **70**.

**[0037]** In exemplary embodiments, heating unit **44** includes one or more heater banks **80**. Each heater bank **80** may be operated as desired to produce heat. In some embodiments, three heater banks **80** may be utilized, as shown. Alternatively, however, any suitable number of heater banks **80** may be utilized. Each heater bank **80** may further include at least one heater coil or coil pass **82**, such as in exemplary embodiments two heater coils or coil passes **82**. Alternatively, other suitable heating elements may be utilized. As is understood, each heater coil pass **82** may be provided as a resistive heating element configured to generate heat in response to resistance to an electrical current flowed therethrough.

**[0038]** As shown, and as would be generally understood in light of the present disclosure, a vent aperture **90** may be defined between the indoor portion **12** and the outdoor portion **14** (e.g., in the rear wall **58** of bulkhead **46**). Vent aperture **90** may allow air flow therethrough between the indoor portion **12** and outdoor portion **14**, and may be utilized in an installed air conditioner unit **10** to allow outdoor air to flow therethrough into the room through the indoor portion **12**. In this regard, in some cases it may be desirable to allow outside air to flow into the room in order to compensate for negative pressure created within the room or indoor environment by, for example, turning on a bathroom fan. In this manner, according to an exemplary embodiment, outside air, also referred to as make-up air, may be provided into the room through vent aperture **90** when a negative pressure is created as air is drawn out of the room by the bathroom fan.

**[0039]** Air conditioner unit **10** may further include one or more auxiliary fans **102** (see FIG. 5) that may be used with the existing refrigeration loop **48** force additional outdoor air through vent aperture **90**. Auxiliary fan **102** may be positioned, for instance, within outdoor portion **14** proximate to vent aperture **90** (e.g., as would be understood). Additionally or alternatively, auxiliary fan **102** may be partially or wholly disposed in vent aperture **90** or partially or wholly disposed in indoor portion **12**. Accordingly, auxiliary fan **102** may induce a flow of outdoor air from the outdoors through vent aperture **90** to the indoor portion **12**. According to the illustrated embodiment auxiliary fan **102** may be controlled by controller **85**, or by any other suitable component.

**[0040]** The operation of air conditioner **10**, including compressor **32** (and thus the sealed system generally), blower fan **42**, fan **33**, heating unit **44**, and other suitable components may be controlled by a control board or controller **85**. Controller **85** may be in communication (via for example a suitable wired or wireless connection) to such components of the air conditioner **10**. By way of example, the controller **85** may include a memory and one or more processing devices such as microprocessors, CPUs or the like, such as general or special purpose microprocessors operable to execute programming instructions or micro-control code associated with operation of air conditioner **10**. The memory may be a separate component from the processor or may be included onboard within the processor. The memory may represent random access memory such as DRAM, or read only memory such as ROM or FLASH. Generally, the processor executes programming instructions stored in memory.

**[0041]** Air conditioner **10** may include a control panel **87** and one or more user inputs **89**, which may be included in control panel **87**. The user inputs **89** may be in communication with the controller **85**. A user of the air conditioner **10** may interact with the user inputs **89** to operate the air conditioner **10**, and user commands may be transmitted between the user inputs **89** and controller **85** to facilitate operation of the air conditioner **10** based on such user commands. A display **88** may additionally be provided in the control panel **87**, and may be in communication with the controller **85**. Display **88** may, for example be a touchscreen or other text-readable display screen, or alternatively may simply be a light that can be activated and deactivated as required to provide an indication of, for example, an event or setting for the air conditioner **10**.

**[0042]** Referring now to FIGS. 1, 4, and 5, a first indoor temperature sensor **92** (e.g., indoor refrigerant temperature sensor) or a second indoor temperature sensor **94** (e.g., indoor ambient temperature sensor) may be disposed within the indoor portion **12**. Each temperature sensor may be configured to sense the temperature of its surroundings. For example, each temperature sensor may be a thermistor or a thermocouple. The indoor temperature sensors **92**, **94** may be in communication with the controller **85**, and may transmit temperatures sensed thereby to the controller **85** (e.g., as one or more voltages or signals, which the controller **85** is configured to interpret as temperature values). Optionally, the voltages or signal transmitted to the controller **85** may be transmitted in response to a polling request or signal received by one or more of the indoor temperature sensors **92**, **94**. For example, a polling request or signal may be transmitted to one or more of the indoor temperature sensors **92**, **94** from the controller **85**.

[0043] First indoor temperature sensor 92 may be disposed proximate the indoor heat exchanger 40 (such as relative to the second indoor temperature sensor 94). For example, in some embodiments, first indoor temperature sensor 92 may be in contact with the indoor heat exchanger 40, such as with a coil 41 thereof. The first indoor temperature sensor 92 may be configured to detect a temperature for the indoor heat exchanger 40. Second indoor temperature sensor 94 may be spaced from the indoor heat exchanger 40, such as in the transverse direction T. For example, the second indoor temperature sensor 94 may be in contact with the room front 24, as illustrated in FIG. 1. Second indoor temperature sensor 94 may be configured to detect a temperature of air entering the indoor portion 12.

[0044] Separate from or in addition to the temperature sensor(s) 92 or 94, some embodiments include a pollutant sensor 126 (e.g., indoor pollutant sensor) (as indicated in phantom lines) that is disposed within the indoor portion 12. Generally, pollutant sensor 126 may be configured to detect the presence, concentration, or level of one or more predetermined suspended pollutants within or relative to the air in the room or indoor environment within which indoor portion 12 is mounted (e.g., in parts per million or one-millionth of a gram of pollutant per gram of air). In other words, pollutant sensor 126 may be configured to detect a pollutant level of the indoor air (i.e., indoor pollutant level). For instance, pollutant sensor 126 may include or be provided as a carbon dioxide (CO<sub>2</sub>) sensor configured to detect the indoor CO<sub>2</sub> level for the indoor environment. Such a CO<sub>2</sub> may include or be provided as a nondispersive infrared sensor, photoacoustic sensor, chemical CO<sub>2</sub> sensor, or other suitable sensor for measuring CO<sub>2</sub> gas. The pollutant sensor 126 may be in communication with the controller 85, and may transmit pollutant levels sensed thereby to the controller 85 (e.g., as one or more voltages or signals, which the controller 85 is configured to interpret as pollutant levels). Optionally, the voltages or signal transmitted to the controller 85 may be transmitted in response to a polling request or signal received by the pollutant sensor 126. For example, a polling request or signal may be transmitted to the pollutant sensor 126 from the controller 85.

[0045] When assembled, the pollutant sensor 126 may be mounted at any suitable portion of unit 10 on or within the indoor portion 12. Optionally, the pollutant sensor 126 may be mounted on or adjacent to the control panel 87. Additionally or alternatively, pollutant sensor may be attached to the air diverter 68, or to another portion of cabinet 20 in fluid communication with the indoor environment.

[0046] Referring especially to FIGS. 1 and 5, in optional embodiments, such as exemplary heat pump unit embodiments, a first outdoor temperature sensor 132 (e.g., outdoor refrigerant temperature sensor) (as indicated in phantom lines) and a second outdoor temperature sensor 134 (e.g., outdoor ambient temperature sensor) (as indicated in phantom lines) are disposed within the outdoor portion 14. Each temperature sensor may be configured to sense the temperature of its surroundings. For example, each temperature sensor may be a thermistor or a thermocouple. The outdoor temperature sensors 132, 134 may be in communication with the controller 85, and may transmit temperatures sensed thereby to the controller 85 (e.g., as one or more voltage signals, which the controller 85 is configured to interpret as temperature readings).

[0047] First outdoor temperature sensor 132 may be disposed proximate the outdoor heat exchanger 30 (such as relative to the second outdoor temperature sensor 134). For example, in some embodiments, first outdoor temperature sensor 132 may be in contact with the outdoor heat exchanger 30, such as with a coil 31 (FIG. 1) thereof. The first outdoor temperature sensor 132 may be configured to detect a temperature for the outdoor heat exchanger 30. Second outdoor temperature sensor 134 may be spaced from the outdoor heat exchanger 30, such as in the transverse direction T. For example, the second outdoor temperature sensor 134 may be in contact with the rear grill 22 (FIG. 1). The second outdoor temperature sensor 134 may be configured to detect a temperature for air entering the outdoor portion 14. During certain operations (e.g., heating operations), air may thus generally flow across or adjacent to the second outdoor temperature sensor 134 and then the first outdoor temperature sensor 132.

[0048] In some embodiments, a remote sensor assembly 210, such as a remote thermostat, is provided at a location separate and apart from the cabinet 20. For instance, the remote sensor assembly 210 may be spaced apart from cabinet 20 while remaining in selective communication with the controller 85 (e.g., via for example a suitable wired or wireless connection). Thus, the remote sensor assembly 210 may be mounted or positioned within the same room as the indoor portion 12, while selectively detecting a temperature that is not immediately adjacent to either the indoor or outdoor portions 12, 14. Additionally or alternatively, the remote sensor assembly 210 may be independently movable relative to the cabinet 20.

[0049] Generally, the remote sensor assembly 210 includes a remote body 212 that houses or supports a suitable sensor circuit 214 for detecting temperature. For instance, the remote sensor assembly 210 may include a sensor circuit 214 that is or includes one or more temperature sensors (e.g., thermocouples, thermistors, optical temperature sensors, infrared temperature sensors, etc.) or pollutant sensors (e.g., nondispersive infrared sensor, photoacoustic sensor, chemical sensor, etc.). Within the remote body 212, a secondary controller 216 may be provided (e.g., in communication with or as part of sensor circuit 214). In additional or alternative embodiments, a network interface 218 may be mounted within the remote body 212 (e.g., to selectively communicate with the controller 85).

[0050] In some embodiments, the secondary controller 216 includes one or more memory devices and one or more processors. The processors of the secondary controller 216 can be any combination of general or special purpose processors, CPUs, or the like that can execute programming instructions or control code associated with operation of remote sensor assembly 210. The memory devices (i.e., memory) of the secondary controller 216 may represent random access memory such as DRAM or read only memory such as ROM or FLASH. In certain embodiments, the processor of the secondary controller 216 executes programming instructions stored in the memory of the secondary controller 216. The memory of the secondary controller 216 may be a separate component from the processor or may be included onboard within the processor. Alternatively, the secondary controller 216 may be constructed without using a processor, for example, using a combination of discrete analog or digital logic circuitry

(such as switches, amplifiers, integrators, comparators, flip-flops, AND gates, and the like) to perform control functionality instead of relying upon software.

**[0051]** In optional embodiments, the secondary controller **216** includes a network interface **218** (e.g., on or off board secondary controller **216**) such that secondary controller **216** can connect to and communicate over one or more networks (e.g., wireless communications network **220**) with the controller **85**. In some such embodiments, network interface **218** includes one or more transmitting, receiving, or transceiving components for transmitting/receiving communications with the controller **85** via wireless communications network **220**. In exemplary embodiments, the wireless communications network **220** may be a wireless sensor network (such as a Bluetooth communication network), a wireless local area network (WLAN), a point-to-point communication networks (such as radio frequency identification networks, near field communications networks, etc.), or a combination of two or more of the above communications networks.

**[0052]** In certain embodiments, the secondary controller **216** is configured to transmit (e.g., wirelessly transmit) one or more detected temperature or pollutant values (i.e., signals corresponding to a value of a temperature or pollutant level detected at remote sensor assembly **210**) to the controller **85**. For example, the secondary controller **216** may be configured to transmit detected temperature or pollutant values unprompted by any outside request, such as a polling request that might otherwise be transmitted to the secondary controller **216** from the controller **85**. Thus, the secondary controller **216** may determine to transmit remote indoor-air (e.g., temperature or pollutant) values independently of the controller **85** or any other device. The receipt of remote indoor values by the controller **85** may be entirely passive or unprompted by the controller **85**. In some such embodiments, the remote indoor values from the secondary controller **216** are transmitted asynchronously or, alternatively, according to a predetermined scheduled (e.g., programmed within the secondary controller **216**). Notably, the lack of a request-polling signal may conserve power (e.g., at the remote sensor assembly **210**) and improve communication between the secondary controller **216** and controller **85**.

**[0053]** Once received, an indoor-air value from the remote sensor assembly **210** (e.g., a temperature value or pollutant level) may be stored (e.g., temporarily) within controller **85**, such as within a temporary or detected field. If the value meets one or more predetermined criteria, the value within the temporary or detected field may be utilized as an operating value, such as an operating temperature (e.g., within an operating temperature field), which the controller **85** may treat as a measure of current condition within a given room (e.g., as the controller **85** directs the sealed system in order to achieve a temperature setpoint provided by a user).

**[0054]** Separate from or in addition to remote sensor assembly **210**, an occupancy reader **250** may be provided. As an example, occupancy reader **250** may be provided on or as part of a remote device, such as within the remote body **212** (e.g., spaced apart from cabinet **20** of unit **10**). As an additional or alternative example, occupancy reader **250** may be mounted directly to or supported on cabinet **20** (e.g., at indoor portion **12**). Generally, occupancy reader **250** is in operative communication with controller **85** and is configured to transmit an occupancy signal thereto. In particular, the occupancy signal transmitted by occupancy reader **250** may indicate the presence or, alternatively, absence of a

person within the same room or indoor environment as unit **10**. For instance, occupancy reader **250** may include an infrared motion sensor, acoustic or ultrasonic sensor, radio frequency sensor, audio sensor, touch sensor, etc. to directly detect a human body within the indoor environment. Thus, a user's directly-detected body may prompt transmission of a direct occupancy signal. Additionally or alternatively, occupancy reader **250** may indirectly detect a human body, such as by including a keycard reader positioned within the indoor environment or on a door thereto. Thus, swiping a mated identification or access card across the occupancy reader **250** may indirectly indicate a user is present within the indoor environment and, thereby, prompt transmission of an indirect occupancy signal.

**[0055]** As an alternative or supplement to occupancy reader **250**, one or more remote computers **240** may be in operative communication with controller **85** and be configured to selectively transmit an occupancy signal thereto. In some such embodiments, the occupancy signal is transmitted from the remote computer **240** based on a remote user input. For instance, the remote computer **240** may be provided be configured as or include a room management device, such as might be provided for a hotel system or network to manage room reservations. In response to a remote user input (e.g., indicating the room has been reserved or an intended occupant has checked in), the remote computer **240** may transmit an occupancy signal indicating the presence of a user within the room or indoor environment of the unit **10**.

**[0056]** Referring now to FIG. 6, the present disclosure may further be directed to methods (e.g., method **600**) of operating an air conditioner or air conditioning appliance, such as air conditioner **10**. In exemplary embodiments, the controller **85** may be operable to perform various steps of a method in accordance with the present disclosure.

**[0057]** The methods (e.g., **600**) may occur as, or as part of, a protective operation (e.g., separate from or in cooperation with a cooling, heating, or dehumidification operation) of the air conditioner **10**. In particular, the methods disclosed herein may advantageously prevent the accumulation of pollutant from the outdoor environment within the indoor environment. Additionally or alternatively, methods disclosed herein may advantageously assess or adept to conditions in the outdoor environment (e.g., pollutant levels) or the indoor environment (e.g., volume of the indoor space) without requiring direct user input, action, or knowledge. Except as otherwise indicated, one or more steps in the below methods (e.g., **600**) may be changed, rearranged, performed in a different order, or otherwise modified without deviating from the scope of the present disclosure.

**[0058]** At **610**, the method **600** includes determining an indoor environment is in an unoccupied state. In other words, it may be determined if the room or indoor environment corresponding to the unit is occupied (e.g., prior to **620**). Occupancy may be determined, for instance, based on an evaluation for a received occupancy signal including information or data from which the current/anticipated presence of one or more human users can be detected. The occupancy signal may include a voltage signal from a sensor which detects a characteristic or condition that has previously been found to indicate a human presence or count (e.g., within the room or indoor environment in which the air conditioner unit is installed). In turn, the absence of an occupancy signal or signal values otherwise indicating no

human presence or count has been detected may be evaluated to determine the indoor environment is in an unoccupied state. Additionally or alternatively, the occupancy signal may include a data packet of user presence (e.g., a binary “present/absent” value or, alternatively, a particular number of users) calculated apart from the air conditioner unit, as is understood. Thus, **610** may include evaluating a received active “absent” signal or otherwise determining the absence of an occupancy signal indicating a user presence.

**[0059]** In some embodiments, **610** includes evaluating for a direct occupancy signal from an external sensor spaced apart from the cabinet of the air conditioner unit (e.g., from an external thermostat or remote occupancy reader), as described above. In additional or alternative embodiments, **610** includes evaluating for an indirect occupancy signal from a remote device spaced apart from the cabinet of the air conditioner unit (e.g., from a remote computer or occupancy reader), as also described above. In further additional or alternative embodiments, **610** includes evaluating for an occupancy signal (e.g., direct occupancy signal or indirect occupancy signal) from a supported occupancy reader attached to the cabinet of the single-package air conditioner unit, as further described above.

**[0060]** At **620**, the method **600** includes detecting a first indoor pollutant level (e.g., following or in response to **610**). For instance, the first indoor pollutant level may be detected from a pollutant sensor within the indoor environment as an initial pollutant level ( $C_0$ ). In some embodiments, **620** includes holding the auxiliary fan in an inactive state (e.g., while detecting the first indoor pollutant level). Separately or as a result of the inactive auxiliary fan, the flow of make-up air may be prevented while detecting the pollutant level.

**[0061]** As described above, one or more pollutant levels may be detected at a pollutant sensor (e.g., carbon dioxide sensor) mounted to the air conditioner unit (e.g., on or within the cabinet). Additionally or alternatively, one or more pollutant levels may be detected at a remote pollutant sensor (e.g., carbon dioxide sensor) mounted apart from the air conditioner unit (e.g., at a remote temperature assembly or thermostat). In turn, the first pollutant level may be detected at a sensor mounted to or apart from the air conditioner unit (e.g., the cabinet thereof) within the indoor environment.

**[0062]** At **630**, the method **600** includes activating an auxiliary fan. Specifically, following **620**, the auxiliary fan may be activated to urge the flow of make-up air to the indoor portion (e.g., from the outdoor portion) following detecting the first indoor pollutant level. Thus, the auxiliary fan may be rotated (e.g., continuously or according to a set program), as would be understood. The airflow or flow rate of the make-up air (e.g., corresponding to rotation speed of the auxiliary fan) may be set in advance. Optionally, the auxiliary fan may continue to be activated or otherwise rotated for the duration of one or more subsequent steps, such as **650**, **660**, or **670**. Additionally or alternatively, activation of the auxiliary fan may maintain a set fan speed or make-up air speed for the duration of activation, such that the volumetric flowrate ( $F_M$ ) (e.g., in cubic feet per minute) of make-up air is known.

**[0063]** At **640**, the method **600** includes detecting a second indoor pollutant level. In particular, the second indoor pollutant level may be detected as a developing pollutant level ( $C_2$ ) following **620** and **630**. For instance, a set or second interval ( $T_2$ ) may be provided following **620** or the

start of **630**. Thus, the second indoor pollutant level may be detected at or in response to expiration of the second interval. Moreover, the second indoor pollutant level may reflect changes in pollutant level within the indoor environment following a set time or duration of make-up air flow. Optionally, the second interval may be between 5 and 15 minutes (e.g., 10 minutes).

**[0064]** In some embodiments, the second pollutant level is detected at the same pollutant sensor as the first pollutant level. In turn, the second pollutant level may be detected at a sensor mounted to or apart from the air conditioner unit (e.g., the cabinet thereof) within the indoor environment.

**[0065]** At **650**, the method **600** includes detecting a third indoor pollutant level. In particular, the third indoor pollutant level may be detected as another developing pollutant level ( $C_3$ ) following **640**. For instance, a set or second interval ( $T_3$ ) may be provided following **620** or the start of **630**. Thus, the third indoor pollutant level may be detected at or in response to expiration of the third interval. Moreover, the third indoor pollutant level may reflect changes in pollutant level within the indoor environment following another set time or duration of make-up air flow. Optionally, the third interval may be between 10 and 30 minutes (e.g., 20 minutes).

**[0066]** In some embodiments, the third pollutant level is detected at the same pollutant sensor as the first or second pollutant level. In turn, the third pollutant level may be detected at a sensor mounted to or apart from the air conditioner unit (e.g., the cabinet thereof) within the indoor environment.

**[0067]** At **660**, the method **600** includes determining one or more concentration characteristics. Specifically, such characteristics may be based on one or more detected values of **620**, **640**, or **650**. In some embodiments, **660** includes determining an outdoor pollutant level ( $C_{out}$ ) based on the first, second, or third indoor pollutant levels. Such a determination may notably be made without directly measuring a pollutant level at the outdoor environment or otherwise outside of the indoor environment. In additional or alternative embodiments, **660** includes determining a room volume ( $V_{room}$ ) of the indoor environment based on the first, second, or third indoor pollutant levels.

**[0068]** For instance, **660** may include determination of outdoor pollutant level ( $C_{out}$ ) or room volume ( $V_{room}$ ) (i.e., one or both of  $C_{out}$  and  $V_{room}$ ) based on one or more programmed characteristic formulas, such as the below:

$$C_2 = (V_{room} - T_2 * F_M) * (C_0 - C_{out}); \text{ and}$$

$$C_3 = (V_{room} - T_3 * F_M) * (C_0 - C_{out});$$

**[0069]** wherein at least  $C_2$ ,  $C_3$ ,  $T_2$ ,  $T_3$ ,  $F_M$ , and  $C_0$  are known (as described above) to solve for  $C_{out}$  or  $V_{room}$ , as would be understood.

**[0070]** At **670**, the method **600** includes estimating a high-pollutant timepoint. In particular, a point or time ( $T_{up}$ ) at which a predetermined upper-limit pollutant level ( $C_{up}$ ) is predicted (e.g., likely to be) reached may be calculated. Such a calculation may be made, for instance, based on one or more characteristics determined at **660**, such as the determined outdoor pollutant level ( $C_{out}$ ) or room volume ( $V_{room}$ ) [e.g., according to one or more programmed formulas, such as  $C_{up} = (V_{room} - T_{up} * F_M) * (C_0 - C_{out})$ ]. Moreover, as would be understood, various further known inputs, such as the initial pollutant level, flowrate of make-up air, etc. may be used to estimate the high-pollutant timepoint.

[0071] At 680, the method 600 includes adjusting activation of the auxiliary fan. For instance, rotation of the auxiliary fan may be reduced or halted (held in an inactive state) prior to or upon reaching the high-pollutant timepoint (e.g., at some predetermined offset interval calculated based on the high-pollutant timepoint). Thus, the method 600 may reduce the flow of make-up air (and thus the rate of pollutant-level increase) before the high-pollutant timepoint is reached. Moreover, 680 may include or continue with holding the auxiliary fan in the inactive state based on the estimated high-pollutant timepoint.

[0072] As would be understood, 680 may continue adjusting or restricting the flow of make-up air until a predetermined return condition is met. Such a predetermined return condition may include detecting an occupancy of the corresponding room or indoor environment, receiving a manual override, detecting a new indoor pollutant level below a predetermined return level, or any other suitable condition.

[0073] This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

What is claimed is:

1. A method of operating a single-package air conditioner unit mounted within an indoor environment, the single-package air conditioner unit comprising a bulkhead defining an indoor portion and an outdoor portion, a vent aperture defined in the bulkhead, and an auxiliary fan for urging a flow of make-up air from the outdoor portion through the vent aperture to the indoor portion, the method comprising:

detecting a first indoor pollutant level within the indoor environment while holding the auxiliary fan in an inactive state;

activating the auxiliary fan to urge the flow of make-up air to the indoor portion following detecting the first indoor pollutant level;

detecting a second indoor pollutant level within the indoor environment following detecting the first indoor pollutant level and activating the auxiliary fan;

determining an outdoor pollutant level based on the first and second indoor pollutant levels;

estimating a high-pollutant timepoint for the indoor environment based on the determined outdoor pollutant level; and

adjusting activation of the auxiliary fan prior to reaching the high-pollutant timepoint.

2. The method of claim 1, further comprising:

detecting a third indoor pollutant level within the indoor environment following detecting the second indoor pollutant level,

wherein determining the outdoor pollutant level is further based on the third indoor pollutant level.

3. The method of claim 1, further comprising:

determining a room volume of the indoor environment based on the first and second indoor pollutant levels.

4. The method of claim 3, further comprising:

detecting a third indoor pollutant level within the indoor environment following detecting the second indoor pollutant level,

wherein determining the room volume is further based on the third indoor pollutant level.

5. The method of claim 1, wherein adjusting activation further comprises holding the auxiliary fan in the inactive state based on the estimated high-pollutant timepoint.

6. The method of claim 1, further comprising:

determining, prior to detecting the first indoor pollutant level, the indoor environment is in an unoccupied state.

7. The method of claim 1, wherein the first and second indoor pollution levels are detected at a carbon dioxide sensor mounted to the single-package air conditioner unit at the indoor portion.

8. The method of claim 1, wherein the first and second indoor pollution levels are detected at a carbon dioxide sensor mounted apart from the indoor portion within the indoor environment.

9. A method of operating a single-package air conditioner unit mounted within an indoor environment, the single-package air conditioner unit comprising a bulkhead defining an indoor portion and an outdoor portion, a vent aperture defined in the bulkhead, and an auxiliary fan for urging a flow of make-up air from the outdoor portion through the vent aperture to the indoor portion, the method comprising:

detecting a first indoor pollutant level within the indoor environment while holding the auxiliary fan in an inactive state;

activating the auxiliary fan to urge the flow of make-up air to the indoor portion following detecting the first indoor pollutant level;

detecting a second indoor pollutant level within the indoor environment following detecting the first indoor pollutant level and activating the auxiliary fan;

determining a room volume of the indoor environment based on the first and second indoor pollutant levels;

estimating a high-pollutant timepoint for the indoor environment based on the determined room volume; and

adjusting activation of the auxiliary fan prior to reaching the high-pollutant timepoint.

10. The method of claim 9, further comprising:

detecting a third indoor pollutant level within the indoor environment following detecting the second indoor pollutant level,

wherein determining the room volume is further based on the third indoor pollutant level.

11. The method of claim 9, wherein adjusting activation further comprises holding the auxiliary fan in the inactive state based on the estimated high-pollutant timepoint.

12. The method of claim 9, further comprising:

determining, prior to detecting the first indoor pollutant level, the indoor environment is in an unoccupied state.

13. The method of claim 9, wherein the first and second indoor pollution levels are detected at a carbon dioxide sensor mounted to the single-package air conditioner unit at the indoor portion.

14. The method of claim 9, wherein the first and second indoor pollution levels are detected at a carbon dioxide sensor mounted apart from the indoor portion within the indoor environment.

**15.** A single-package air conditioner unit mounted within an indoor environment, the single-package air conditioner unit comprising:

- a cabinet defining an outdoor portion and an indoor portion;
- an outdoor heat exchanger disposed in the outdoor portion;
- an indoor heat exchanger disposed in the indoor portion;
- a compressor in fluid communication with the outdoor heat exchanger and the indoor heat exchanger to circulate a refrigerant between the outdoor heat exchanger and the indoor heat exchanger;
- a bulkhead disposed between the indoor portion and the outdoor portion, the bulkhead defining a vent aperture extending therethrough to permit airflow between the indoor portion and the outdoor portion;
- an auxiliary fan mounted within the cabinet in fluid communication with the vent aperture for urging a flow of make-up air from the outdoor portion through the vent aperture to the indoor portion; and
- a controller in operative communication with the compressor and the auxiliary fan, the controller being configured to initiate a conditioning operation, the conditioning operation comprising
  - detecting a first indoor pollutant level within the indoor environment while holding the auxiliary fan in an inactive state,
  - activating the auxiliary fan to urge the flow of make-up air to the indoor portion following detecting the first indoor pollutant level,
  - detecting a second indoor pollutant level within the indoor environment following detecting the first indoor pollutant level and activating the auxiliary fan,
  - determining an outdoor pollutant level and a room volume based on the first and second indoor pollutant levels,

- estimating a high-pollutant timepoint for the indoor environment based on the determined outdoor pollutant level and the room volume, and
- adjusting activation of the auxiliary fan prior to reaching the high-pollutant timepoint.

**16.** The single-package air conditioner unit of claim **15**, wherein the conditioning operation further comprises

- detecting a third indoor pollutant level within the indoor environment following detecting the second indoor pollutant level,

- wherein determining the outdoor pollutant level and the room volume is further based on the third indoor pollutant level.

**17.** The single-package air conditioner unit of claim **15**, wherein adjusting activation further comprises holding the auxiliary fan in the inactive state based on the estimated high-pollutant timepoint.

**18.** The single-package air conditioner unit of claim **15**, wherein the conditioning operation further comprises

- determining, prior to detecting the first indoor pollutant level, the indoor environment is in an unoccupied state.

**19.** The single-package air conditioner unit of claim **15**, further comprising:

- a carbon dioxide sensor mounted to the single-package air conditioner unit at the indoor portion,

- wherein the first and second indoor pollution levels are detected at the carbon dioxide sensor.

**20.** The single-package air conditioner unit of claim **15**, further comprising:

- a carbon dioxide sensor mounted apart from the indoor portion within the indoor environment,

- wherein the first and second indoor pollution levels are detected at the carbon dioxide sensor.

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